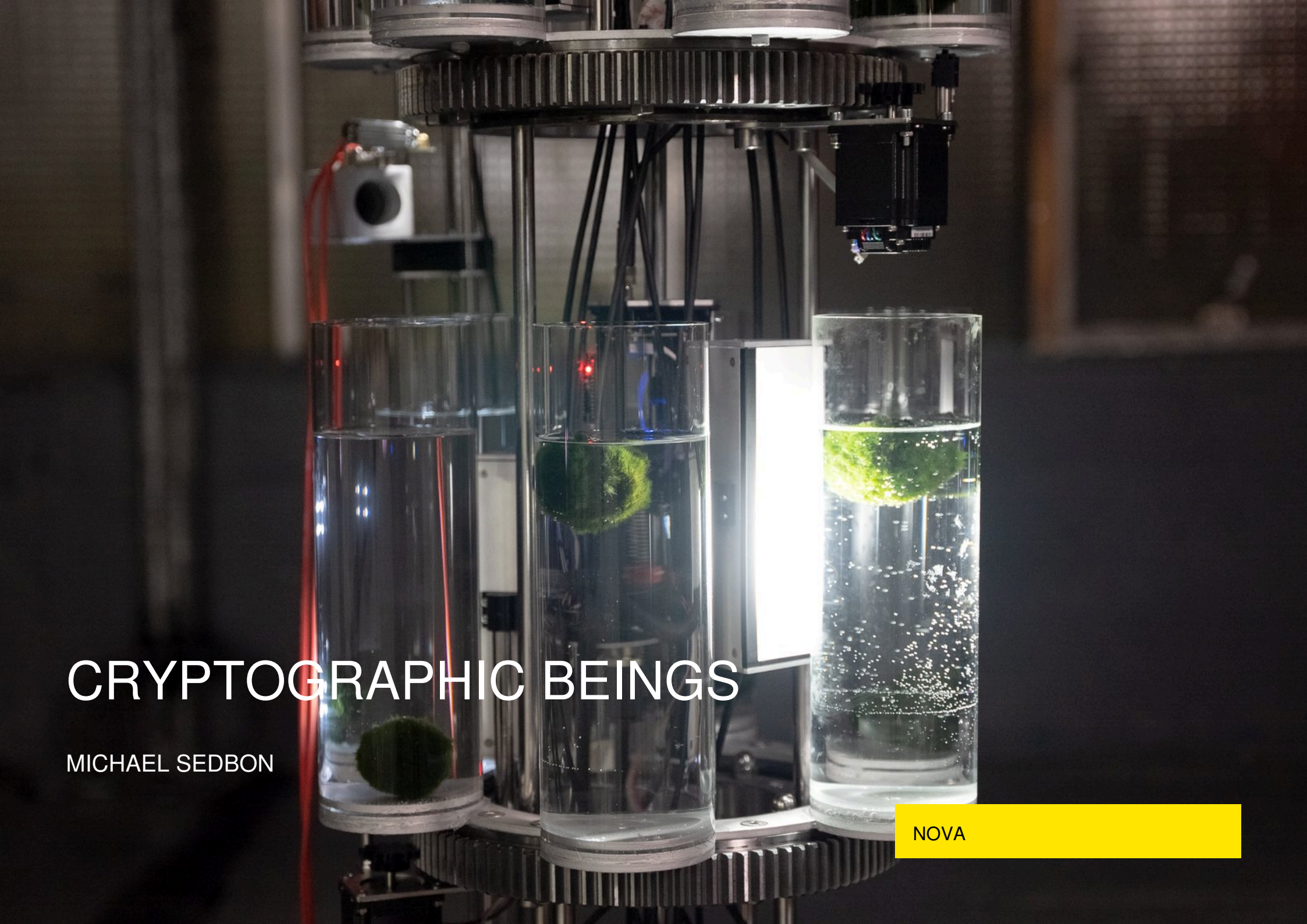

CRYPTOGRAPHIC BEINGS

MICHAEL SEDBON

NOVA

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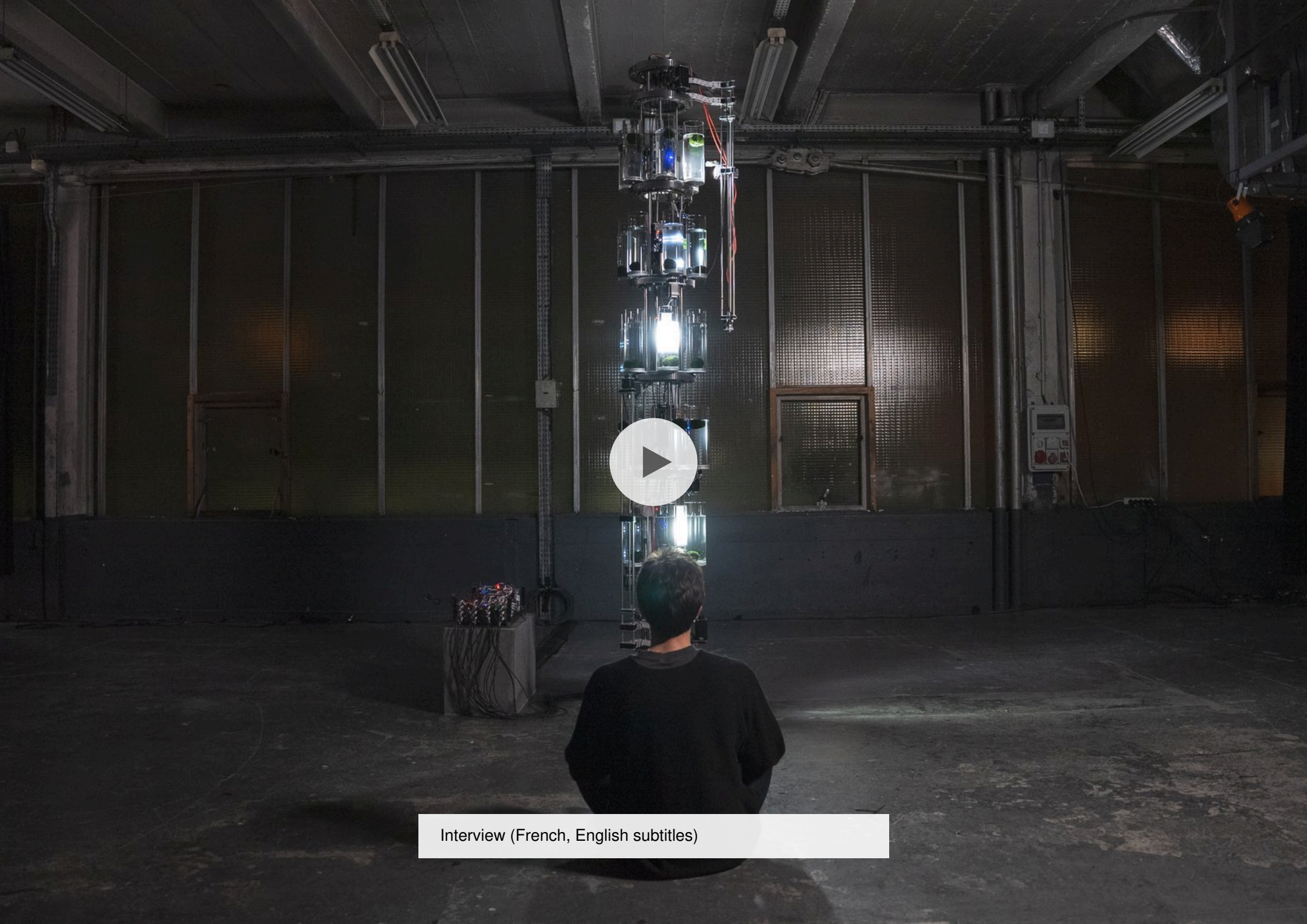
CRYPTOGRAPHIC BEINGS

Custom bio reactors & software, motors, kinetic lights, Aegagropila linnaei

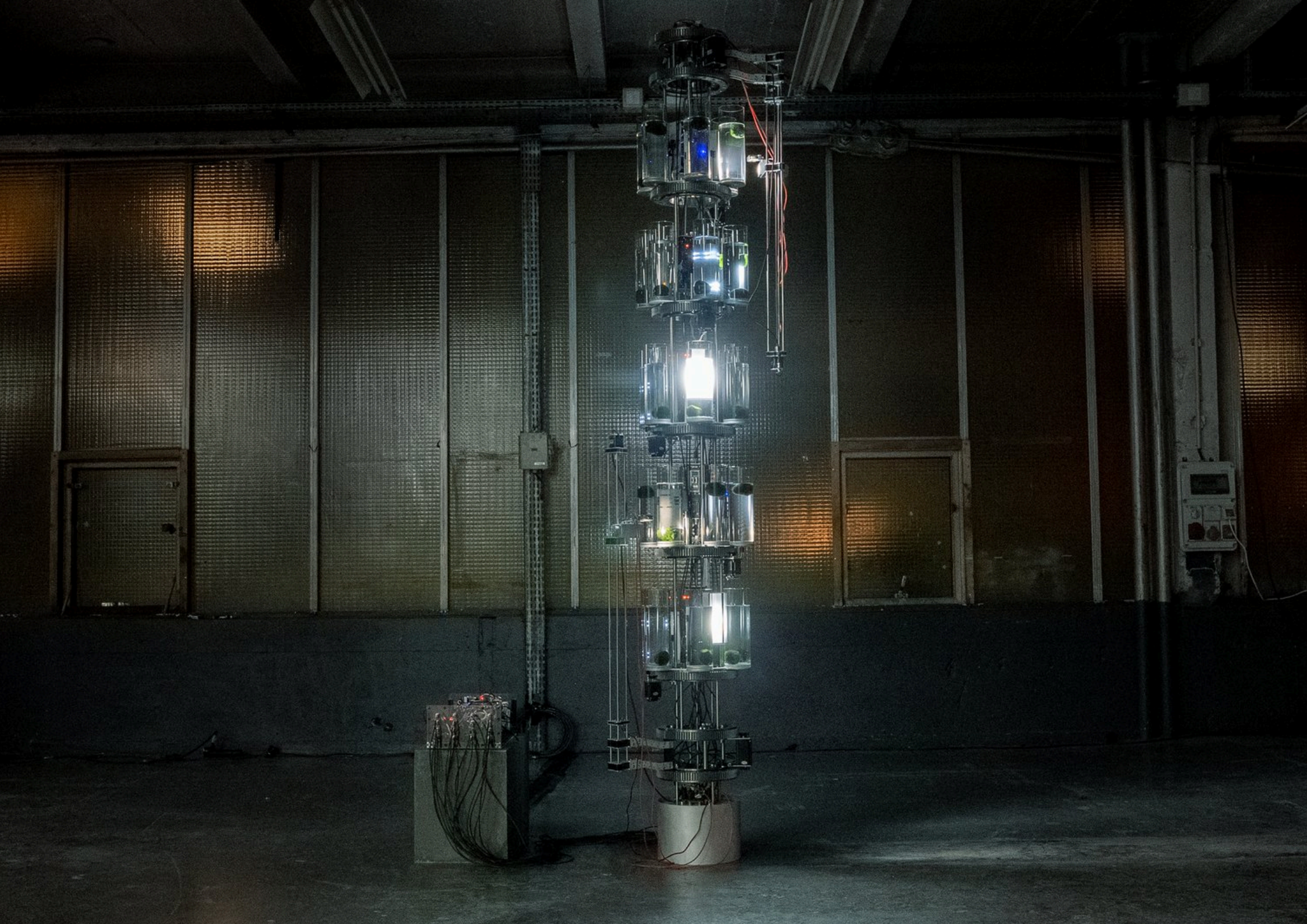
Cryptographic Beings is a technological proposal for a living hard-drive where digital information is stored in living (vegetal) media instead of silicon based transistors. It calls for a redefinition of our relationship to both living organisms and man-made technologies.

The project realises binary logic through photosynthetic buoyancy switching: each living bit is a marimo algae sphere (Aegagropila linnaei). When illuminated, photosynthesis supersaturates the sphere with oxygen; micro-bubbles, visible to the naked eye, become trapped inside its filaments, decreasing its density so the sphere rises and floats. In darkness the oxygen diffuses away and the sphere gradually sinks.

In this setup, data durability now depends on biological health, so every write cycle consumes real metabolic resources: light, nutrients, recovery time.



Interview (French, English subtitles)



NOVA Utopia <> Dystopia

Positioning within the curatorial question

NOVA asks whether AI will guide us toward a glittering UTOPIA or pull us into a dark DYSTOPIA.

Cryptographic Beings deliberately inhabits the fault-line between those extremes. By encoding digital information inside living algae, the project turns biology into an active memory substrate. Each write operation costs energy, nutrients and recovery time; permanence is achievable only through care.

Because the installation makes both trajectories viscerally tangible, it aligns with NOVA insistence that the time to wonder is over. Viewers must decide whether to cultivate or consume.

A custom Algorithm for NOVA

A reprogrammable living machine

Cryptographic Beings is not a fixed sculpture but a re-programmable living machine: a grid of marimo-algae bits whose chlorophyll can be flashed with binary light patterns to store and later reveal data.

For NOVA I am proposing to rewrite its control code so that, over the exhibition three months, the installation itself becomes a slow, self-modifying experiment.

Each day a language model writes a short passage into the living array, then reads back two simple signals: 1) did the bits survive? 2) are the algae still photosynthesising after the write?

MICHAEL SEDBON

Bio Hybrid computation Design Studio

Head of Cyber Synthetic Biology @ OXMAN NYC

Michael Sedbon is an interaction designer, artist and life science researcher currently based in New York.

His work explores digital networked technologies and systems through their convergence with non-human intelligence (plants, unicellular organisms, insects, bacterias etc...) in regards to the Infocene problematics.

He explores how cultural practices around artificial intelligence intersect with the natural phenomena of biological intelligence by building bio-computers that manipulate living organisms through rules encoded in software, hardware, and wetware.

Michael has been the recipient of the Bio Arts and Design Award and a Falling Walls winner while his work has been recognised in several international prizes such as the STARTS prize of the European Commission and the Lumens Prize.

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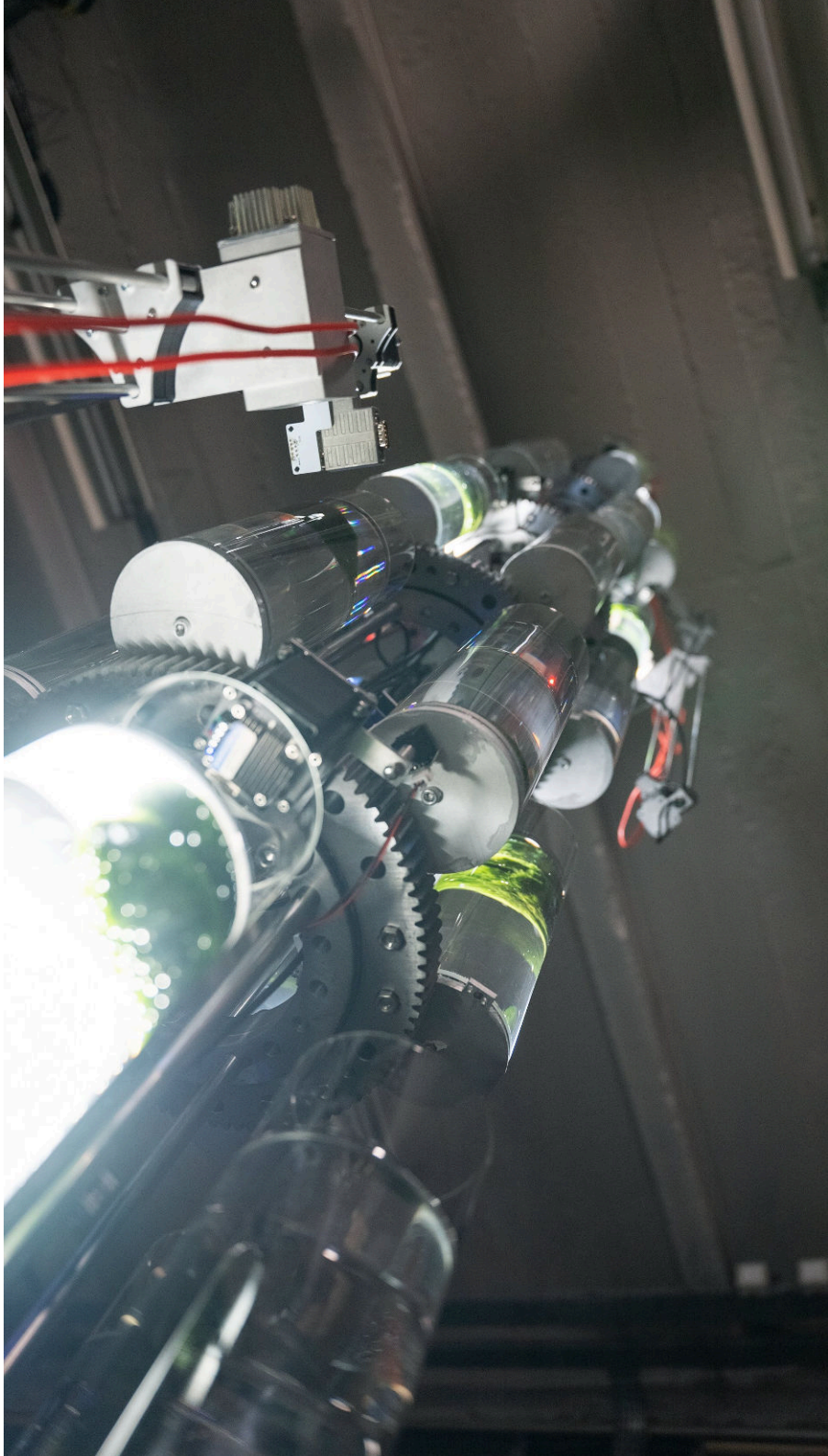
CRYPTOGRAPHIC BEINGS

TECHNICAL RIDER

TECHNICAL RIDER

Space Required	::	On site: 6 m2 floor area x 4 m height
Electricity	::	Three 220V plug
Installation	::	2 Days on site
Dismantling	::	2 Days on site
Start-up	::	Single automated start-up routine with a single button
Transportation	::	Size: 9m3